



Technical Presentation

Title: From Caesar to Vigenère: Evolution of Substitution Ciphers

Course name: Cryptography and Network Security

Course code: CSA5116

Name : Siripireddy Hari Chandana

Reg no: 192372136



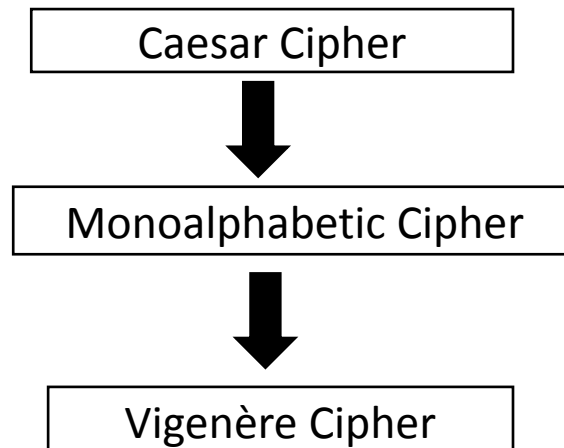
INTRODUCTION

From Caesar to Vigenère: Evolution of Substitution Ciphers

What is a Substitution Cipher?

- A classical encryption technique where each plaintext letter is replaced with another letter.
- The goal is to protect information from unauthorized access.
- The security of the cipher depends on the substitution rule (or key).

Evolution of Substitution Ciphers





Caesar Cipher

How it Works

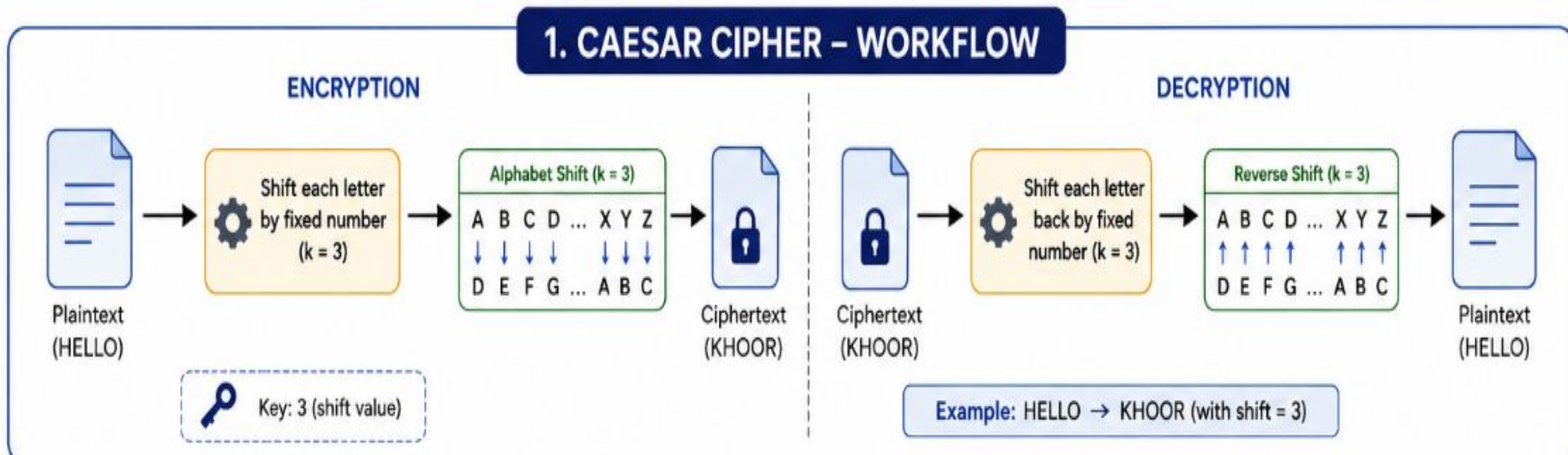
- Invented by **Julius Caesar** around **50 BC**.
- Each letter is shifted by a fixed number of positions.
- Commonly uses a shift of **3**.

Example

Plaintext	H	E	L	L	O
Shift (+3)	K	H	O	O	R

Ciphertext: KHOOR

Caesar Cipher





Monoalphabetic Cipher

How it Works

- Each plaintext letter is replaced with a unique, randomly assigned letter.
- The same substitution is used throughout the message.

Example Mapping

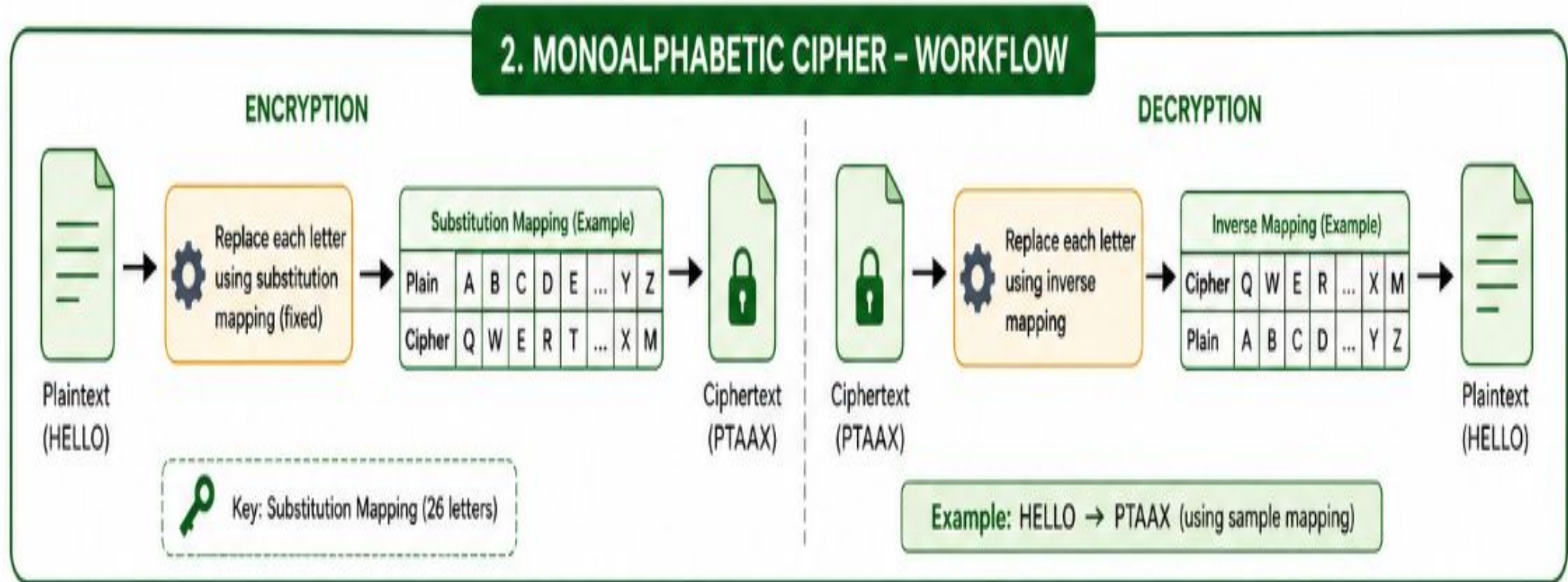
Plain	A	B	C	D	E	F
Cipher	Q	W	E	R	T	Y

Example

Plaintext: HELLO

Ciphertext: PTAAX (*Example mapping*)

Monoalphabetic Cipher





Vigenère Cipher

How it Works

- Uses a **keyword** to perform multiple Caesar shifts.
- The shift changes for each letter, making encryption much stronger.

Example

- **Plaintext:** HELLO
- **Keyword:** KEYKE

Plain	H	E	L	L	O
Key	K	E	Y	K	E
Cipher	R	I	J	V	S

Ciphertext: RIJVS

Vigenère Cipher

3. VIGENÈRE CIPHER - WORKFLOW

ENCRYPTION



Plaintext
(HELLO)



Use keyword to
apply different
shifts to each letter

Position	1	2	3	4	5
Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E
Shift (A=0)	10	4	24	10	4
Ciphertext	R	I	J	V	S



Ciphertext
(RIJVS)



Key: Keyword (e.g., KEY)

DECRYPTION



Ciphertext
(RIJVS)



Use keyword to
apply reverse
shifts

Position	1	2	3	4	5
Ciphertext	R	I	J	V	S
Keyword	K	E	Y	K	E
Shift (A=0)	10	4	24	10	4
Plaintext	H	E	L	L	O



Plaintext
(HELLO)

Example: HELLO + KEY → RIJVS



*Thank
you!*